磁电转换特性

线圈电流/mA	300	250	200	150	100	50	0
磁感应强度B/Gs	5.973	4.977	3.982	2.986	1.991	0.995	0
输出电压/V	1.500	1.270	1.029	0.780	0.528	0.264	0.002
线圈电流/mA	-50	-100	-150	-200	-250	-300	
磁感应强度B/Gs	-0.995	-1.991	-2.986	-3.982	-4.977	-5.973	
输出电压/V	-0.264	-0.522	-0.777	-1.029	-1.273	-1.502	

各向异性特性

夹角/°	0	10	20	30	40	50	60	70	80	90
输出电压/V	1.027	1.014	0.973	0.906	0.809	0.688	0.541	0.378	0.198	0.009

轴向磁场测量

位置	-0.5R	-0.4R	-0.3R	-0.2R	-0.1R	0	0.1R	0.2R	0.3R	0.4R	0.5R
V(x)测量值/V	0.966	0.995	1.012	1.019	1.021	1.023	1.022	1.021	1.017	1.002	0.975
B(x)测量值/Gs	3.782	3.895	3.962	3.989	3.997	4.005	4.001	3.997	3.981	3.923	3.817
B(x)/B(0)测量值	0.944	0.973	0.989	0.996	0.998	1.000	0.999	0.998	0.994	0.979	0.953
B(x)/B(0)理论值	0.946	0.975	0.992	0.998	1.000	1.000	1.000	0.998	0.992	0.975	0.946
相对误差	0.211%	0.205%	0.302%	0.200%	0.200%	0.000%	0.100%	0.000%	0.202%	0.410%	0.740%

空间磁场测量（原数据：输出电压U/V）

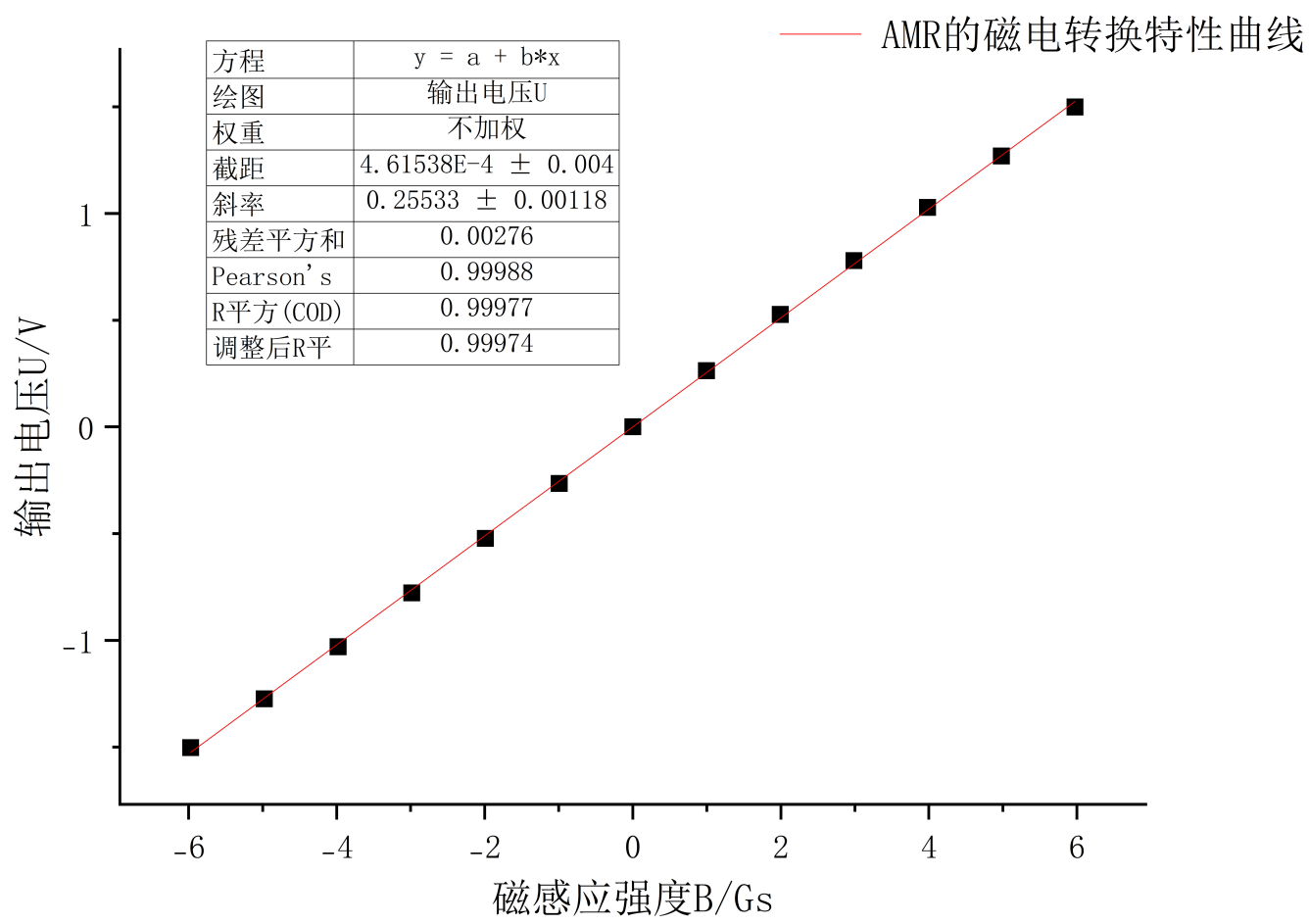
Y轴 \ X轴	0	0.05R	0.10R	0.15R	0.20R	0.25R	0.30R
0	1.020	1.023	1.022	1.022	1.020	1.018	1.013
0.05R	1.020	1.023	1.022	1.022	1.020	1.018	1.014
0.10R	1.020	1.022	1.022	1.022	1.021	1.019	1.015
0.15R	1.020	1.022	1.022	1.022	1.022	1.021	1.018
0.20R	1.019	1.021	1.022	1.023	1.024	1.024	1.022
0.25R	1.018	1.020	1.021	1.023	1.026	1.028	1.028
0.30R	1.015	1.018	1.020	1.024	1.028	1.032	1.034

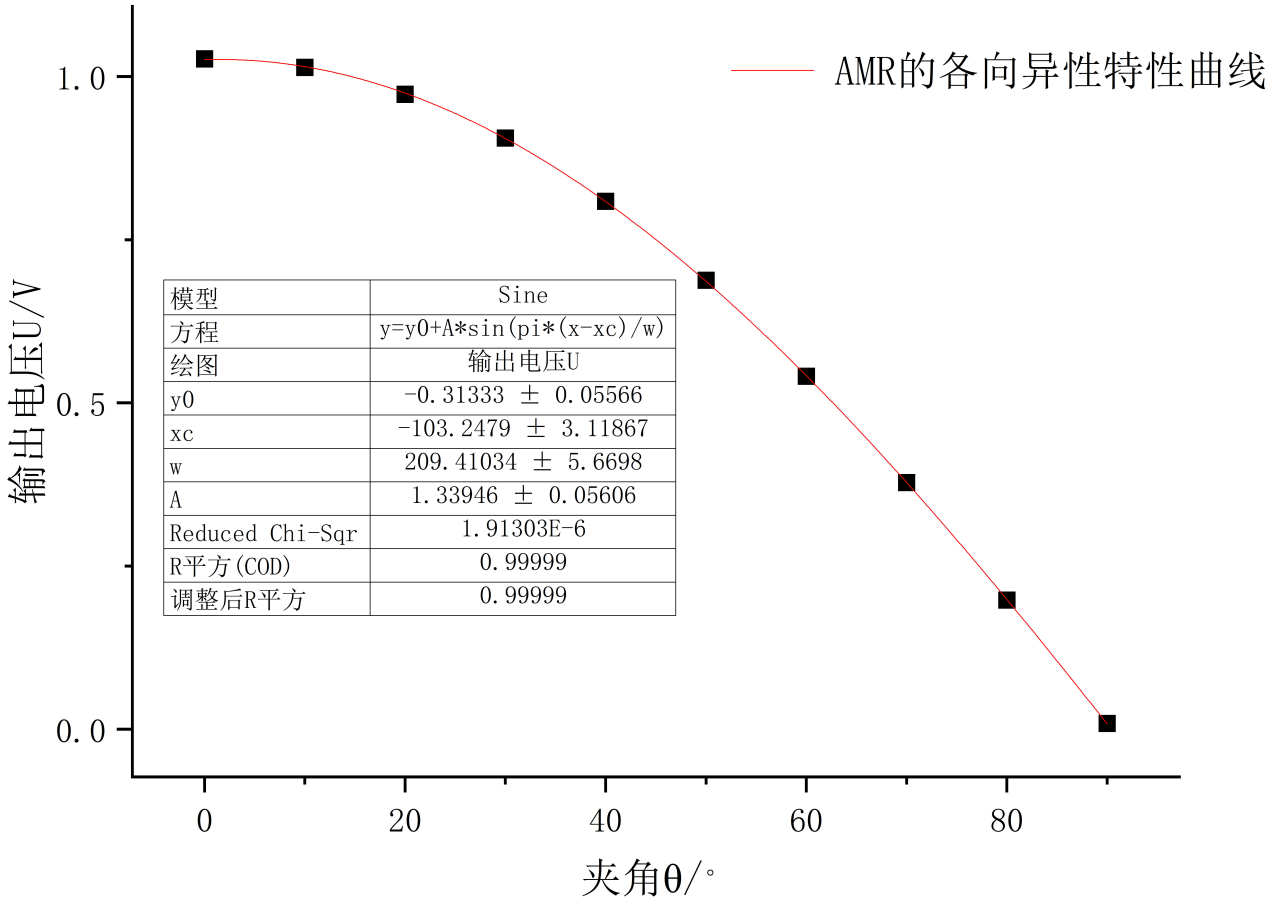
空间磁场测量（数据处理：磁感应强度B/Gs）

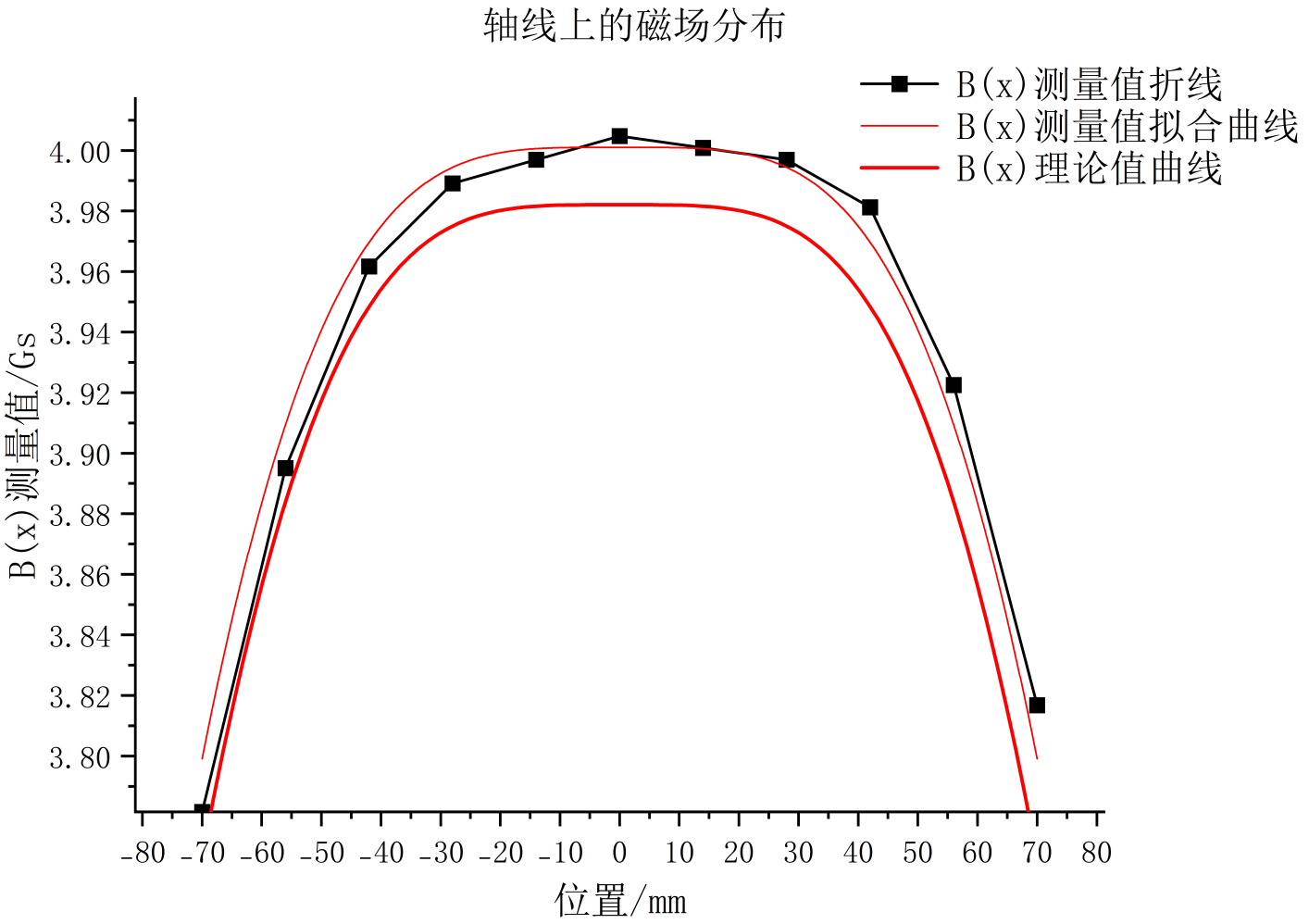
Y轴(mm) \ X轴(mm)	0	7	14	21	28	35	42
0	3.993	4.005	4.001	4.001	3.993	3.985	3.967
7	3.993	4.005	4.001	4.001	3.993	3.985	3.970
14	3.993	4.001	4.001	4.001	3.997	3.989	3.973
21	3.993	4.001	4.001	4.001	4.001	3.997	3.985
28	3.989	3.997	4.001	4.005	4.009	4.009	4.001
35	3.985	3.993	3.997	4.005	4.017	4.024	4.024
42	3.973	3.985	3.993	4.009	4.024	4.040	4.048

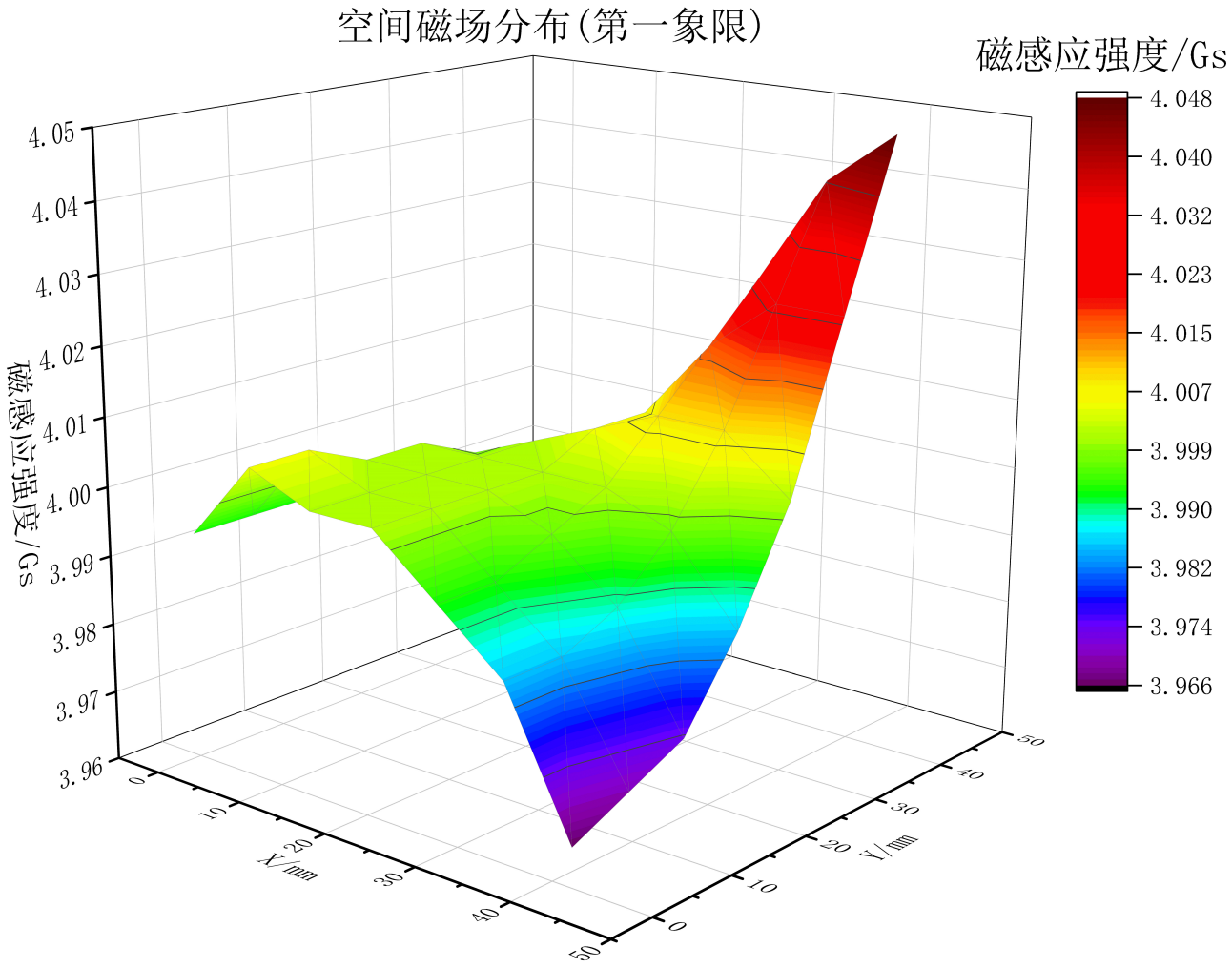
地磁场测量

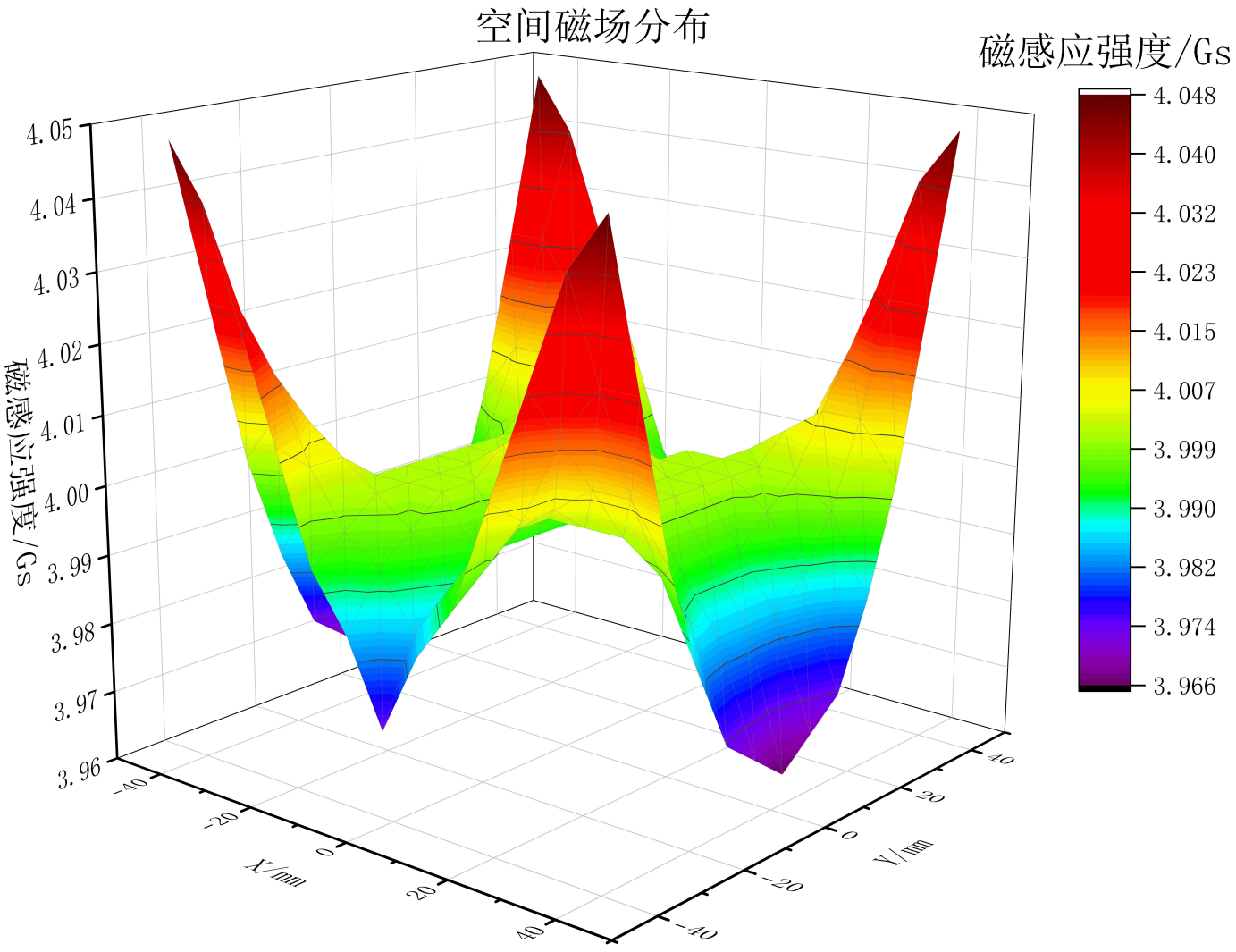
磁偏角D/°	0
磁倾角I/°	41
输出电压U_1/V	0.099
输出电压U_2/V	-0.127
$U=(U_1-U_2)/2$ /V	0.113
B=U/0.25 /Gs	0.452


AMR的磁电转换特性曲线
方程 y = a + b*x
绘图 输出电压U
权重 不加权
截距 4.61538E-4 ± 0.004
斜率 0.25533 ± 0.00118
残差平方和 0.00276
Pearson's 0.99988
R平方(COD) 0.99977
调整后R平 0.99974
输出电压U/V
1
0
-1
-6
-4
-2
0
2
4
6
磁感应强度B/Gs


AMR的各向异性特性曲线
模型 Sine
方程 y=y0+A*sin(pi*(x-xc)/w)
绘图 输出电压U
y0 -0.31333 ± 0.05566
xc -103.2479 ± 3.11867
w 209.41034 ± 5.6698
A 1.33946 ± 0.05606
Reduced Chi-Sqr 1.91303E-6
R平方(COD) 0.99999
调整后R平方 0.99999
输出电压U/V
1.0
0.5
0.0
0
20
40
60
80
夹角θ/°


轴线上的磁场分布
B(x)测量值折线
B(x)测量值拟合曲线
B(x)理论值曲线
B(x)测量值/Gs
4.00
3.98
3.96
3.94
3.92
3.90
3.88
3.86
3.84
3.82
3.80
-80 -70 -60 -50 -40 -30 -20 -10 0 10 20 30 40 50 60 70 80
位置/mm


空间磁场分布(第一象限)
磁感应强度/Gs
4.048
4.040
4.032
4.023
4.015
4.007
3.999
3.990
3.982
3.974
3.966
磁感应强度/Gs
4.05
4.04
4.03
4.02
4.01
4.00
3.99
3.98
3.97
3.96
X/mm
Y/mm


空间磁场分布
磁感应强度/Gs
4.048
4.040
4.032
4.023
4.015
4.007
3.999
3.990
3.982
3.974
3.966
磁感应强度/Gs
4.05
4.04
4.03
4.02
4.01
4.00
3.99
3.98
3.97
3.96
-40
-20
0
20
40
X/mm
Y/mm